

College	RV COLLEGE OF ENGINEERING®			
Department	COMPUTER SCIENCE AND ENGINEERING			
Program:	M.Tech in CSE	Major Project	Course Code	MCE442P
Student Name	Veerasagar S S		USN	1RV24SCS17
Project Title	ACTIS: An Agentic Collaborative Threat Intelligence System using Trust-Aware Federated Learning			
Under taken at	RV COLLEGE OF ENGINEERING			
Internal Guide	Dr. Rajashree Shettar		Designation - Professor	
External Guide	-		-	

SYNOPSIS

Introduction: As cyber threats become increasingly sophisticated, organizations face a critical dilemma: they must share threat intelligence to defend against zero-day attacks, but are restricted from sharing raw network data due to strict data privacy regulations and the risk of exposing Personally Identifiable Information (PII). Current collaborative security architectures either rely on centralized data pooling which introduces privacy risks and Single Points of Failure (SPOFs)—or suffer from isolated defense mechanisms where an attack pattern known to one node is entirely unknown to others. Fed-Intel is proposed to resolve this by employing a dual-channel federated architecture. It combines Federated Learning (FL) to collaboratively train Intrusion Detection Systems (IDS) without exposing data, and Federated Retrieval-Augmented Generation (RAG) integrated with an Agentic Privacy Engine to safely share actionable zero-day threat intelligence.

Objectives:

- Develop a Collaborative IDS: Implement a federated learning framework using a Trust-Aware FedAvg aggregation strategy to improve local IDS accuracy across non-IID datasets without centralizing raw traffic.
- Ensure Absolute Data Privacy: Design an LLM-powered Agentic Privacy Engine to automatically sanitize network flow anomalies and guarantee zero PII leakage (e.g., stripping IPs, MACs, credentials).
- Enable Cross-Organizational Intelligence Sharing: Build a Global Federated RAG Knowledge Base to store and semantically search safely summarized threat intelligence.
- Automate Proactive Defenses: Implement an Immunity Engine that queries the global RAG database to auto-generate and deploy perimeter defenses.

Methodology:

Data Pipeline & FL: Different enterprise and datasets (NF-CSE-CIC-IDS2018-v2, NF-BoT-IoT-v2) are distributed across multiple simulated nodes. Local PyTorch-based Multilayer Perceptron (MLP) models are trained. Only the model weights and local loss metrics are transmitted to a central FL Server.

Trust-Aware Aggregation: The global server blends a sample-weighted average with a trust-based weight (inversely proportional to local loss) to generate and distribute an updated global model, penalizing underperforming or malicious nodes.

Agentic Privacy Pipeline: When an anomaly is detected, a LangChain-orchestrated agent (utilizing Gemini 2.0 Flash) analyzes the attack, maps it to MITRE ATT&CK tactics, and strips all sensitive PII. A Differential Privacy (DP) layer applies calibrated Gaussian noise to the threat's embedding.

Federated RAG & Immunity: Sanitized semantic embeddings are synchronized with a ChromaDB vector store. The RAG pipeline retrieves relevant attack profiles, enabling the LLM-driven Immunity Engine to translate cross-client threat intelligence into actionable firewall configurations.

Software Requirements:

- Programming Language: Python 3.10+
- Machine Learning & FL: PyTorch 2.0+, Scikit-learn, Numpy, Pandas
- Agentic AI & NLP: LangChain, langchain-google-genai (Google Gemini 2.0 API), sentence-transformers
- Vector Database: ChromaDB
- Interfaces & IPC: Typer, Rich (Terminal Dashboard UI), ZeroMQ

Hardware Requirements:

- Development: Laptop/Workstation (16GB+ RAM, SSD, CPU with A VX).
- Training: GPU instance (NVIDIA T4/RTX series or equivalent) recommended for faster training
- Inference: CPU inference possible with quantized models; GPU recommended
- Storage: 100 GB (logs, datasets, models); Network access for containers.

Innovation / Contributions:

The gap none of these papers fill: Active PII sanitization (stripping IPs, payloads, user data from threat summaries before sharing). This is Fed-Intel's unique contribution, the agentic LLM that reads logs and outputs privacy-safe summaries.

Internal Guide	Signature of Dean CSE-Cluster	Signature of the HOD
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